Conditions for Oscillation in the Wilson-Cowan Model

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Abstract

We investigate why the Neuromatch Academy (NMA) default parameters for the Wilson-Cowan model cannot produce oscillations under any external input. Starting from the unstable requirements on the Jacobian trace and determinant, we derive a structural necessary condition on the network parameters: $w_{EI}w_{IE} > 16\tau_E w_{II}/(\tau_I a_E)$. This single inequality determines whether oscillation is possible regardless of external input or fixed-point tuning. The NMA parameters violate it ($52 < 73.3$); the oscillatory parameters of Li et al. (2022) satisfy it with wide margin ($520 \gg 32$). By coupling the sigmoid derivatives along the I-nullcline, we tighten this bound and show that the decoupled optimum is an exact special case of the coupled problem. An I-nullcline sweep algorithm over $N = 5000$ random parameter sets confirms zero theoretical violations and establishes that a unique unstable fixed point is the essential divide: 97.8% oscillation versus 0.3% for multi-fixed-point systems. In the end we discussed our current limitations and future work.

In computational neuroscience, the **Wilson-Cowan model** (Hugh R. Wilson & Jack D. Cowan) describes the dynamics of interactions between populations of very simple excitatory and inhibitory model neurons. It is a mean-field model that tracks the **proportion of active neurons** in each population over time.

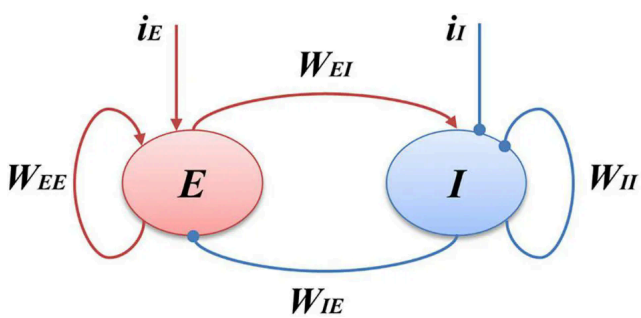

Figure 1: Schematic of the Wilson-Cowan model[1]

$$\tau_E \frac{dr_E}{dt} = -r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) \tag{1}$$

$$\tau_I \frac{dr_I}{dt} = -r_I + F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \tag{2}$$

where the activation function F is a sigmoid:

[1] E represents the excitatory populations, I stands for the inhibitory populations, the red arrows indicate the excitatory projections and blue arrows show the inhibitory projections, i_I is the external inputs of the I populations, and i_E is the external inputs of the E populations. Li et al. (2022)

$$F(x; a, \theta) = \frac{1}{1 + e^{-a(x-\theta)}} - \frac{1}{1 + e^{a\theta}} \tag{3}$$

The correction term $-\frac{1}{1+e^{a\theta}}$ ensures $F(0) = 0$ (no input yields no activation).

> **Remark**: We use the simplified Wilson-Cowan equations without the refractory term $(1 - r_E)$, following the formulation in Neuromatch Academy.

0.1 Variables

- $r_E(t)$ — average firing rate (or proportion of active neurons) of the **excitatory** population at time t. Ranges from 0 (silent) to 1 (maximally active).
- $r_I(t)$ — average firing rate of the **inhibitory** population at time t.

0.2 Parameters

0.2.1 Time Constants

- τ_E — time constant of the excitatory population.
- τ_I — time constant of the inhibitory population.

0.2.2 Sigmoid Parameters

- a_E, a_I — gain (steepness) of the activation function for E and I populations. Higher gain means a sharper threshold-like response.
- θ_E, θ_I — threshold of the activation function. The input level at which the population reaches half-maximal activation.

0.2.3 Connection Weights

- w_{EE} — excitatory $\longrightarrow$ excitatory coupling strength (recurrent excitation).
- w_{EI} — inhibitory $\longrightarrow$ excitatory coupling strength (how much I suppresses E).
- w_{IE} — excitatory $\longrightarrow$ inhibitory coupling strength (how much E drives I).
- w_{II} — inhibitory $\longrightarrow$ inhibitory coupling strength (recurrent inhibition).

0.2.4 External Inputs

- I_E^{ext} — external input current to the excitatory population (primary control parameter for bifurcation analysis).
- I_I^{ext} — external input current to the inhibitory population (typically fixed at 0).

0.2.5 Parameter Values

Parameter	Value	Description
τ_E	1.0 ms	Average synaptic time constant for excitatory population
τ_I	2.0 ms	Average synaptic time constant for inhibitory population
a_E	1.2	Excitatory gain
a_I	1.0	Inhibitory gain
θ_E	2.8	Excitatory threshold
θ_I	4.0	Inhibitory threshold
w_{EE}	9.0	Strength of self-excitatory feedback
w_{EI}	4.0	Strength of connections between E–I
w_{IE}	13.0	Strength of connections between I–E
w_{II}	11.0	Strength of self-inhibitory feedback
I_E^{ext}	0.0	External input to excitatory population (sweep this)
I_I^{ext}	0.0	External input to inhibitory population

Table 1: Default parameter values from the Neuromatch Academy computational neuroscience curriculum.

Remark: The parameters play an important role in the system, please read the *Periodic solution (Oscillations)* part in details.

1 Nullcline Analysis

1.1 E-nullcline

Let $\dot{r}_E = 0$, then

$$\frac{1}{\tau_E}(-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}})) = 0 \tag{4}$$

$$-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = 0 \tag{5}$$

$$F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = r_E \tag{6}$$

Then we use the inverse of F (namely the logit function):

$$F^{-1}(y; a, \theta) = \theta + \frac{1}{a} \ln \frac{y + c}{1 - y - c}, \quad c = \frac{1}{1 + e^{a\theta}} \tag{7}$$

Applying F_E^{-1} to both sides:

$$F_E^{-1}(r_E) = w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}} \tag{8}$$

Solving for r_I:

$$r_I = \frac{1}{w_{EI}} (w_{EE} \cdot r_E - F_E^{-1}(r_E) + I_E^{\text{ext}}) \tag{9}$$

1.2 I-nullcline

Analogously, let $\dot{r}_I = 0$, we have

$$r_I = F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \tag{10}$$

$$r_E = \frac{1}{w_{IE}} (w_{II} \cdot r_I + F_I^{-1}(r_I) - I_I^{\text{ext}}) \tag{11}$$

1.3 Phase plane

After substituting the parameters, we obtain the figure

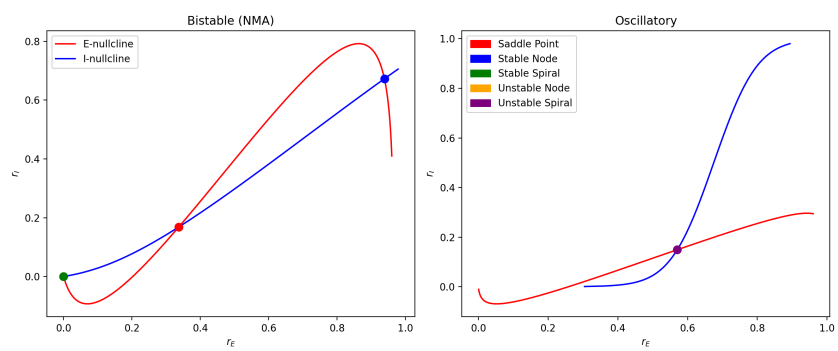

Figure 2: Phase plane analysis

Inserting point $(0.1, 0.5)$, it's easy to check the arrow direction. We omit here.

> **Remark:** The domains are $r_E \in (0, 0.9664)$ and $r_I \in (0, 0.9820)$, as the logarithm requires its argument to be positive.

2 Qualitative Analysis

Note that in the previous section, we obtain the phase plane with 3 equilibrium points. Now we use the Jacobian matrix to determine its state. Recall the Jacobian matrix:

$$J = \begin{pmatrix} \frac{\partial \dot{r}_E}{\partial r_E} & \frac{\partial \dot{r}_E}{\partial r_I} \\ \frac{\partial \dot{r}_I}{\partial r_E} & \frac{\partial \dot{r}_I}{\partial r_I} \end{pmatrix} \tag{12}$$

We solve the partial derivative and obtain

$$J = \begin{pmatrix} \frac{1}{\tau_E}(-1 + w_{EE} \cdot F'_E) & \frac{1}{\tau_E}(-w_{EI} \cdot F'_E) \\ \frac{1}{\tau_I}(w_{IE} \cdot F'_I) & \frac{1}{\tau_I}(-1 - w_{II} \cdot F'_I) \end{pmatrix} \tag{13}$$

in which

$$F'(x) = a \cdot (F(x) + c)(1 - F(x) - c), \quad c = \frac{1}{1 + e^{a\theta}} \tag{14}$$

Note that F' is evaluated at the sigmoid input, not at r_E or r_I directly:

$$u_E = w_{EE} r_E - w_{EI} r_I + I_E^{\text{ext}}, \quad u_I = w_{IE} r_E - w_{II} r_I + I_I^{\text{ext}} \tag{15}$$

We classify each fixed point using the trace $\tau = \text{tr}(J)$, determinant $\Delta = \det(J)$, and discriminant $\tau^2 - 4\Delta$ of the Jacobian:

- $\Delta < 0 \Rightarrow$ saddle point
- $\Delta > 0, \tau < 0 \Rightarrow$ stable (node if $\tau^2 - 4\Delta > 0$, spiral if < 0)
- $\Delta > 0, \tau > 0 \Rightarrow$ unstable

For our 3 fixed points:

Fixed Point	τ	Δ	$\tau^2 - 4\Delta$	**Classification**
$(0, 0)$	-1.247	0.406	-0.069	Stable Spiral
$(\approx 0.34, 0.17)$	$-$	-0.941	$-$	Saddle Point
$(\approx 0.94, 0.68)$	-2.403	1.381	0.250	Stable Node

Table 2: Fixed point classification for default parameters ($I_E^{\text{ext}} = 0$).

The system with default parameters ($I_E^{\text{ext}} = 0$) is therefore **bistable**. (In a dynamical system, **bistability** means the system has two stable equilibrium states a bistable structure can be resting in either of two states.)

3 Periodic Solutions (Oscillations)

Recall the **Poincaré-Bendixson theorem** states that for a two-dimensional system, for which we know that the solutions will stay in **a bounded area** after some finite time and there is **no fixed point**, there must exist a periodic solution to which the trajectories converge.

We sweep the I_E^{ext} until we obtain only one fixed point which is unstable.

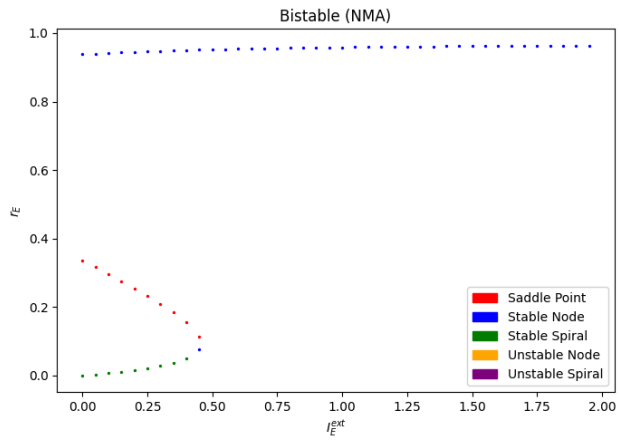

Figure 3: Stable fixed point output after sweeping I_E^{ext}

Note that after sweeping I_E^{ext}, we only get the stable 1 fixed point, why?

The answer lies in the network's structural parameters.

Comparing with the parameters used in Li et al. (2022) (Table 1, based on Wilson and Cowan (1972) and Jadi and Sejnowski (2014)) for gamma oscillation research:

Parameter	**NMA (bistability)**	**Table 1 (oscillation)**
w_{EE}	9	16
w_{EI}	4	26
w_{IE}	13	20
w_{II}	11	1
τ_E (ms)	1	20
τ_I (ms)	2	10
θ_E, θ_I	2.8, 4.0	5, 20
$I_E^{\text{ext}}, I_I^{\text{ext}}$	0, 0	2, 7
a_E, a_I	1.2, 1.0	1.0, 1.0

Table 3: Parameter comparison: bistable vs. oscillatory regime.

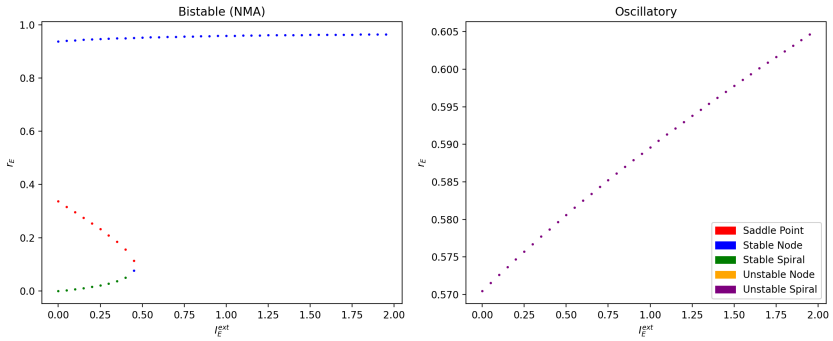

Figure 4: Output comparison between NMA (bistable) and Li et al. (oscillatory) parameter regimes.

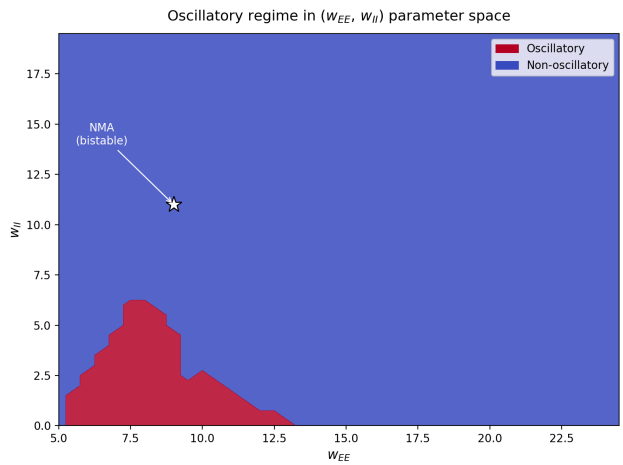

Figure 5: Oscillatory regime in (Wee, WI) parameter space for NMA parameters

3.1 Conditions for Oscillation

Our goal is to find the limit cycle. Based on Poincaré-Bendixson theorem as we mentioned, the system is bounded by sigmiod. In addition, Wilson-Cowan has at least one fixed point. So we require it to be unstable. The cleanest case is: having only one unstable fixed point(tr > 0, det > 0). In this section we derive conditions on the structural parameters under which this can be achieved.

An unstable spiral — the state immediately beyond a Hopf bifurcation — requires a fixed point where the Jacobian has both positive trace (instability) and positive determinant (spiral, rather than saddle):

$$\tau = \frac{1}{\tau_E}(-1 + w_{EE}F_E') + \frac{1}{\tau_I}(-1 - w_{II}F_I') \tag{16}$$

$$\Delta = \frac{1}{\tau_E\tau_I}[1 + w_{II}F_I' - w_{EE}F_E' + (w_{EI}w_{IE} - w_{EE}w_{II})F_E'F_I'] \tag{17}$$

- $\tau > 0$: In order to let $\frac{1}{\tau_E}(-1 + w_{EE}F_E') + \frac{1}{\tau_I}(-1 - w_{II}F_I') > 0$, we need

$$w_{EE}F_E' > 1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F_I') \tag{18}$$

- $\Delta > 0$: In order to let

$$\frac{1}{\tau_E \tau_I}[1 + w_{II}F_I' - w_{EE}F_E' + (w_{EI}w_{IE} - w_{EE}w_{II})F_E'F_I'] > 0 \tag{19}$$

  we require

$$1 + w_{II}F_I' + w_{EI}w_{IE}F_E'F_I' > w_{EE}F_E' + w_{EE}w_{II}F_E'F_I' \tag{20}$$

$$1 + w_{II}F_I' + w_{EI}w_{IE}F_E'F_I' > w_{EE}F_E'(1 + w_{II}F_I') \tag{21}$$

  Rearranging as an upper bound on $w_{EE}F_E'$:

$$w_{EE}F_E' < \frac{1 + w_{II}F_I' + w_{EI}w_{IE}F_E'F_I'}{1 + w_{II}F_I'} \tag{22}$$

Combining Equation 18 and Equation 22, a feasibility window for $w_{EE}F_E'$ exists only when the lower bound is less than the upper bound:

$$1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F_I') < \frac{1 + w_{II}F_I' + w_{EI}w_{IE}F_E'F_I'}{1 + w_{II}F_I'} \tag{23}$$

Multiplying both sides by $(1 + w_{II}F_I')$ and simplifying:

$$w_{EI}w_{IE} > \frac{\tau_E}{\tau_I} \cdot \frac{(1 + w_{II}F_I')^2}{F_E'F_I'} \tag{24}$$

Since F_E' and F_I' are not free parameters but determined by the fixed point, we minimise the right-hand side to obtain the weakest necessary condition.

Maximising F_E'. F_E' appears only in the denominator, so we need its maximum. From Equation 14:

$$F'(x) = a \cdot (F(x) + c)(1 - F(x) - c) \tag{25}$$

This is maximised when $F(x) + c = 1/2$, i.e. at the sigmoid midpoint $u_E = \theta_E$. Substituting:

$$F_E'^{\max} = a_E \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{a_E}{4} \tag{26}$$

Minimising over F_I'. Let $\varphi = F_I' > 0$. The right-hand side of Equation 24 becomes:

$$R(\varphi) = \frac{\tau_E}{\tau_I} \cdot \frac{(1 + w_{II}\varphi)^2}{(a_E/4) \cdot \varphi} = \frac{4\tau_E}{\tau_I a_E} \cdot \frac{1 + 2w_{II}\varphi + w_{II}^2\varphi^2}{\varphi} \tag{27}$$

Expanding:

$$R(\varphi) = \frac{4\tau_E}{\tau_I a_E}\left(\frac{1}{\varphi} + 2w_{II} + w_{II}^2\varphi\right) \tag{28}$$

Differentiating and setting to zero:

$$R'(\varphi) = \frac{4\tau_E}{\tau_I a_E}\left(-\frac{1}{\varphi^2} + w_{II}^2\right) = 0 \tag{29}$$

$$\varphi^2 = \frac{1}{w_{II}^2} \quad \Longrightarrow \quad \varphi^* = \frac{1}{w_{II}} \tag{30}$$

To confirm this is a minimum:

$$R''(\varphi) = \frac{4\tau_E}{\tau_I a_E} \cdot \frac{2}{\varphi^3} > 0 \quad \text{for all } \varphi > 0 \tag{31}$$

Substituting $F_E' = a_E/4$ and $F_I' = 1/w_{II}$:

$$R(\varphi^*) = \frac{4\tau_E}{\tau_I a_E}(w_{II} + 2w_{II} + w_{II}) = \frac{4\tau_E}{\tau_I a_E} \cdot 4w_{II} = \frac{16\tau_E w_{II}}{\tau_I a_E} \tag{32}$$

This yields the structural necessary condition[2]:

$$w_{EI} w_{IE} > \frac{16\tau_E w_{II}}{\tau_I a_E} \tag{33}$$

Due to the F_E' and F_I' are jointly determined by the fixed-point coordinates, they cannot in general simultaneously attain their optimal values. This makes Equation 33 a potentially loose bound. See Equation 40 for a tightened bound.

Note that the condition means that the Hopf bifurcation is possibly reachable in this parameter space.

This is a purely structural necessary condition: if it fails, no fixed point can undergo Hopf bifurcation regardless of where it sits on the sigmoid. We verify against both parameter sets:

- **Li et al.**: $\frac{16\times20\times1}{10\times1} = 32$, and $w_{EI}w_{IE} = 520 \gg 32$. ✓
- **NMA**: $\frac{16\times1\times11}{2\times1.2} = 73.3$, and $w_{EI}w_{IE} = 52 < 73.3$. ×

Note that for Li et al. parameters, $w_{II} = 1 < \frac{4}{a_I} = 4$, so the optimal $F_I' = \frac{1}{w_{II}} = 1$ exceeds the sigmoid maximum $\frac{a_I}{4} = 0.25$. The unconstrained bound gives 32; the constrained bound (using $F_I' = \frac{a_I}{4}$) gives $\frac{16\times20\times(1+1\times0.25)^2}{10\times1\times1} = 50$. Since $520 \gg 50$, the conclusion is unchanged, but for parameter sets with $w_{EI}w_{IE}$ between 32 and 50, only the tighter bound is reliable.

The role of each parameter in Equation 24 is:

- $w_{EI}w_{IE}$ (left-hand side): the cross-population coupling is the sole resource available to meet the cost. It represents the strength of the $E \to I \to E$ feedback loop.
- τ_E/τ_I: the time-constant ratio sets the baseline cost. When $\tau_E > \tau_I$ (inhibition recruited faster, the "weakly modulated" regime of Keeley et al. (2019)), the cost is higher.
- $(1 + w_{II}F_I')^2$: inhibitory self-feedback raises the cost quadratically.
- F_E' (denominator only): the excitatory population must operate near its sensitive region ($u_E \approx \theta_E$) to maximise F_E' and keep the cost manageable.
- F_I' (both numerator and denominator): increasing F_I' lowers the cost through the denominator but raises it through $(1 + w_{II}F_I')^2$ in the numerator. The optimal balance is at $F_I' = 1/w_{II}$.

[2]Indeed, the condition could be tightened to $w_{EI}w_{IE} > \frac{16\tau_E(1+w_{II}a_I/4)^2}{\tau_I a_E a_I}$ in this regime(see section 4).

> From Keeley et al. (2019): "...weakly modulated network gamma is found when the firing rate of the inhibitory population is recruited faster than that of the excitatory population, and strongly synchronized gamma oscillations are seen for the reverse. In both frameworks, network oscillations arise when a steady state loses stability via Hopf bifurcation..."

An immediate corollary concerns the cross-term $w_{EI}w_{IE}$ vs $w_{EE}w_{II}$. When $\tau_E > \tau_I$, assume $w_{EI}w_{IE} \leq w_{EE}w_{II}$. Then Equation 21 requires $1 + w_{II}F_I' > w_{EE}F_E'$. Combined with Equation 18:

$$1 + w_{II}F_I' > 1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F_I') \tag{34}$$

Dividing by $(1 + w_{II}F_I') > 0$ gives $\tau_E < \tau_I$, a contradiction. Therefore, in the weakly modulated regime, $w_{EI}w_{IE} > w_{EE}w_{II}$ is strictly necessary.

3.1.1 From necessity to sufficiency

The structural condition assumes both sigmoid derivatives simultaneously take their most favourable values. In practice, the fixed-point position determines the actual F_E' and F_I', and the full master inequality (Equation 24) must be satisfied at a realised fixed point. This introduces two additional requirements:

- **Trace bound** ($\tau > 0$): $w_{EE}F_E'$ must exceed the threshold in Equation 18. This requires w_{EE} to be large enough to place u_E near θ_E, keeping F_E' near its maximum. But if w_{EE} is too large, saturation collapses F_E' and the trace condition fails from the other direction. The trace thus imposes both a lower and an upper bound on w_{EE} through the behaviour of F_E'.
- **Determinant upper bound** ($\Delta > 0$): independently, $w_{EE}F_E'$ must not exceed the algebraic ceiling in Equation 22. This bound can be violated even when F_E' is still large enough for $\tau > 0$ so that the determinant turns negative while the trace remains positive, producing a Saddle.

Together with the structural condition, these requirements fully characterise the oscillatory regime at three levels:

1. **Structural** (Equation 33): can the network support oscillation at all? A necessary condition on coupling weights and time constants alone.
2. **Window exists** (Equation 24 at realised fixed point): does the feasibility window $L < U$ remain open at the actual sigmoid slopes? Necessary but not sufficient.
3. **Fixed point in window** ($L < w_{EE}F_E' < U$): does the actual $w_{EE}F_E'$ fall between the trace lower bound and the determinant upper bound? This guarantees the fixed point is an unstable spiral. **A limit cycle additionally requires that no other stable attractor captures the trajectories**, which is satisfied when this is the system's only fixed point[3].

[3] We discussed it in section 3.2.3.

3.1.2 Tightening via the I-nullcline

The structural condition (Equation 33) treats F_E' and F_I' as independent. In the actual system they are coupled through the fixed-point coordinates. We now exploit the I-nullcline geometry to reduce the two-dimensional optimisation to a single variable.

Key observation. The I-nullcline

$$r_I = F_I(w_{IE} r_E - w_{II} r_I + I_I^{\text{ext}}) \tag{35}$$

is independent of I_E^{ext}. Changing I_E^{ext} shifts the E-nullcline but leaves the I-nullcline unchanged, so the fixed point slides along the I-nullcline. Both sigmoid derivatives therefore become functions of a single coordinate:

$$\alpha(r_E) = F_E'(r_E), \quad \beta(r_E) = F_I'(r_I(r_E)) \tag{36}$$

and the master inequality (Equation 24) right-hand side becomes:

$$R(r_E) = \frac{\tau_E}{\tau_I} \cdot \frac{(1 + w_{II}\beta)^2}{\alpha \cdot \beta} \tag{37}$$

Logarithmic differentiation. To find the minimum of R, we differentiate $\ln R$:

$$\ln R = \text{const} + 2\ln(1 + w_{II}\beta) - \ln\alpha - \ln\beta \tag{38}$$

$$\frac{d\ln R}{dr_E} = \frac{2w_{II}\beta'}{1 + w_{II}\beta} - \frac{\alpha'}{\alpha} - \frac{\beta'}{\beta} = 0 \tag{39}$$

where $\alpha' = d\alpha/dr_E$ and $\beta' = d\beta/dr_E$. Collecting the β' terms in the stationarity condition:

$$\beta' \cdot \frac{w_{II}\beta - 1}{\beta(1 + w_{II}\beta)} = \frac{\alpha'}{\alpha} \tag{40}$$

To evaluate β', we differentiate the I-nullcline implicitly:

$$\frac{dr_I}{dr_E} = \frac{w_{IE}\beta}{1 + w_{II}\beta} \tag{41}$$

so that

$$\beta'(r_E) = a_I(1 - 2r_I - 2c_I) \cdot \frac{w_{IE}\beta}{1 + w_{II}\beta} \tag{42}$$

The decoupled optimum as a special case. When $\beta = 1/w_{II}$, the left-hand side of Equation 40 vanishes identically, forcing $\alpha' = 0$, i.e. $F_E' = a_E/4$. This is precisely the decoupled optimum used to derive Equation 33. The structural bound is therefore not an approximation but an exact stationary point of the coupled problem that may or may not lie on the I-nullcline.

When is the bound tight? If the I-nullcline passes through the point where $\beta = 1/w_{II}$ and simultaneously $\alpha = a_E/4$, then Equation 33 is exact. When $w_{II} < 4/a_I$, the optimal $F_I' = 1/w_{II}$ exceeds the sigmoid maximum $a_I/4$; the true minimum of R is attained at the boundary, making Equation 33 looser. Moreover, the sign of $(w_{II}\beta - 1)$ controls the trade-off direction: when $\beta < 1/w_{II}$ the two derivatives compensate in opposite directions along the I-nullcline; when $\beta > 1/w_{II}$ they move together.

Explicit parameterisation. Using the second form of the I-nullcline (Equation 11):

$$r_{E(r_I)} = \frac{1}{w_{IE}}(w_{II} r_I + F_I^{-1}(r_I) - I_I^{\text{ext}}) \tag{43}$$

both derivatives become explicit functions of r_I:

$$\beta(r_I) = a_I(r_I + c_I)(1 - r_I - c_I) \tag{44}$$

$$\alpha(r_I) = a_E\Big(r_{E(r_I)} + c_E\Big)\Big(1 - r_{E(r_I)} - c_E\Big) \tag{45}$$

so that $R(r_I)$ is fully explicit:

$$R(r_I) = \frac{\tau_E}{\tau_I} \cdot \frac{(1 + w_{II}\beta(r_I))^2}{\alpha(r_I) \cdot \beta(r_I)} \tag{46}$$

The I_E^{ext} that places a fixed point at $\Big(r_{E(r_I)}, r_I\Big)$ is obtained by inverting the E-nullcline:

$$I_E^{\text{ext}} = F_E^{-1}(r_E) - w_{EE} r_E + w_{EI} r_I \tag{47}$$

The minimum of $R(r_I)$ has no closed form but requires only a single numerical sweep over r_I with no root-finding.

Necessary condition. If $\min_{r_I} R(r_I) > w_{EI} w_{IE}$, then no I_E^{ext} can open the feasibility window at any point on the I-nullcline, and oscillation is impossible. This is strictly tighter than Equation 33.

Almost sufficient. If $\min_{r_I} R(r_I) < w_{EI} w_{IE}$, then some I_E^{ext} produces an unstable fixed point. However, at that I_E^{ext} the E-nullcline may intersect the I-nullcline at three points, and the competing stable attractor suppresses limit cycle formation. The sufficient condition additionally requires that I_E^{ext} lies in the one-fixed-point region, discussed in Section 5.

3.2 Verification[4]

3.2.1 Structural necessary condition

We first check Equation 33 against both parameter sets:

	NMA	**Li et al.**
$w_{EI} w_{IE}$	52	520
$\frac{16\tau_E w_{II}}{\tau_I a_E}$	73.3	32
Result	$52 < 73.3$ ×	$520 \gg 32$ ✓

Table 4: Structural necessary condition (Equation 33). NMA fails even under the most favourable sigmoid slopes, ruling out Hopf bifurcation for any w_{EE} or I_E^{ext}.

For Li et al., the condition is satisfied with wide margin, and the system indeed oscillates at the default parameters ($u_E = 5.24 \approx \theta_E = 5.0$, confirming that w_{EE} is tuned to keep F_E' near its maximum).

[4]To verify it, use our condition test module.

3.2.2 Fixed points

We verify across the (w_{EE}, w_{II}) parameter space using NMA cross-coupling values ($w_{EI} = 4$, $w_{IE} = 13$):

w_{EE}	w_{II}	I_E^{ext}	Structural	Window	In window	Observed behaviour
11	0	0.50	✓	✓	✓	**Unstable Spiral** (oscillation)
11	1	0.50	✓	✓	✓	**Unstable Node** (oscillation)
11	5	0.00	✓	×	×	Stable Spiral + Saddle + Stable Node
14	2	0.00	✓	✓	×	Stable Spiral + Saddle
20	0	0.00	✓	×	×	Stable Node + Saddle
9	11	0.00	×	×	×	Stable Spiral + Saddle + Stable Node

Table 5: Three-level verification of oscillation conditions.[5]

Each failure mode corresponds to a different level:

- $(w_{EE} = 9, w_{II} = 11)$ (NMA defaults): fails at level 1. The structural condition gives $52 < 73.3$, ruling out oscillation regardless of w_{EE} tuning or external input.
- $(w_{EE} = 11, w_{II} = 5)$: passes level 1 ($52 > 33.3$), but fails at level 2. Increasing w_{II} shifts the fixed points so that the actual F_E' and F_I' deviate from their optimal values, collapsing the feasibility window.
- $(w_{EE} = 20, w_{II} = 0)$: passes level 1 ($52 > 0$), but fails at level 2. Saturation pushes $F_E' \approx 0$ at both fixed points, inflating the right-hand side of Equation 24 far beyond $w_{EI} w_{IE} = 52$.
- $(w_{EE} = 14, w_{II} = 2)$: passes level 1 ($52 > 13.3$) and level 2, the window exists. But fails at level 3: the saddle achieves $\tau > 0$ ($w_{EE} F_E'$ exceeds the lower bound), yet $w_{EE} F_E'$ also exceeds the algebraic ceiling from Equation 22, pushing $\Delta < 0$. The actual operating point overshoots the window from above.
- $(w_{EE} = 11, w_{II} = 0)$ and $(11, 1)$: all three levels satisfied. w_{EE} places u_E near θ_E, keeping F_E' near-maximal, and the actual $w_{EE} F_E'$ falls within the feasibility window.

3.2.3 Monte Carlo validation

To test the framework beyond hand-picked parameters, we sampled 5000 random parameter combinations uniformly across wide ranges ($w_{EE}, w_{EI}, w_{IE} \in [1, 30]$, $w_{II} \in [0, 20]$, $\tau_E, \tau_I \in$

[5] **Structural**: Equation 33 satisfied.
Window: feasibility window $L < U$ exists at the realised fixed point (Equation 24).
In window: actual $w_{EE} F_E'$ falls between trace lower bound and determinant upper bound.
Oscillation requires all three.

$[0.5, 30]$, $a_E, a_I \in [0.5, 3]$, $\theta_E \in [1, 10]$, $\theta_I \in [1, 20]$, $I_E^{\text{ext}}, I_I^{\text{ext}} \in [-2, 5]$) and compared the analytical predictions against numerical integration with oscillation detection.

Category	Count	Oscillating	Rate
Level 1 fail	1698	0	0.00%
Level 1 pass, Level 2 fail	2613	1	0.04%
Level 1–2 pass, Level 3 fail	351	15	4.27%
Level 3 pass	338	142	42.01%

Table 6: Monte Carlo validation ($N = 5000$)[6].

The structural necessary condition (Level 1) is fully confirmed: none of the 1698 samples that failed it produced oscillation. The single Level 2 anomaly (1 out of 2613) has parameters $w_{EE} = 18.8$, $w_{II} = 13.6$, $\tau_E = 5.1$, $\tau_I = 14.4$; the system has a single stable fixed point (no unstable point). We classify this as a probable false positive[7] ; definitive confirmation would require computing the first Lyapunov coefficient(we omit here).

Similarly, all 15 middle-zone oscillations involve systems with no unstable fixed point. The same caveat applies: subcritical Hopf cannot be excluded without Lyapunov coefficient analysis, but the oscillation characteristics are consistent with detector false positives.

The Level 3 confirmation rate of 42% reveals an important refinement. Level 3 guarantees that a fixed point is locally unstable (Hopf condition), but not that a limit cycle forms. In multi-fixed-point systems, an unstable spiral can coexist with stable attractors that capture trajectories before a limit cycle develops. To verify this interpretation, we partition the Level 3 passes by fixed-point count:

Fixed-point structure	Count	Oscillating	Rate
1 fixed point (unstable)	145	130	89.7%
3 fixed points	162	10	6.2%
Other	31	2	6.5%

Table 7: Level 3 passes partitioned by fixed-point count.[8]

The 10 oscillating three-fixed-point cases all share the classification U1/S1/Sad1 (one unstable spiral, one stable attractor, one saddle). No U2/S0/Sad1 configuration, in which a global limit cycle could enclose all three fixed points, was observed.

Across all 5000 samples, zero genuine violations of the analytical framework were observed. The complete sufficient condition for oscillation in the Wilson-Cowan model is therefore: Level 3 satisfied at the system's unique fixed point.

When a single fixed point exists and is an unstable spiral, the system is bounded ($r_E, r_I \in$

[6]Oscillation detected via numerical integration (RK4) with periodicity analysis after transient removal.

[7]While a stable limit cycle can in principle coexist with a stable fixed point via subcritical Hopf bifurcation, this would require the first Lyapunov coefficient to be positive, which is a scenario not ruled out analytically but unlikely given that the detected "oscillation" has characteristics consistent with a slowly decaying spiral near the Hopf boundary.

[8]Systems with a unique unstable fixed point oscillate at $\approx 90\%$; the gap from 100% is attributable to numerical limitations (fixed-point detection and oscillation sensitivity). Multi-fixed-point systems rarely oscillate despite passing Level 3, confirming that competing stable attractors suppress limit cycle formation.

$[0, 1]$)[9]and contains no other attractor, so the Poincaré-Bendixson theorem guarantees a limit cycle.

3.2.4 I-nullcline sweep

The analysis in Section 3.1.2 provides a concrete algorithm using the explicit parameterisation by r_I:

1. Compute $\min_{r_I} R(r_I)$. If $\min_{r_I} R(r_I) > w_{EI} w_{IE}$, oscillation is impossible for any I_E^{ext} (tighter necessary condition than Equation 33).
2. Identify the feasible interval: all r_I where $\tau > 0$ and $\Delta > 0$ (the Hopf region on the I-nullcline).
3. For each r_I in the Hopf region, invert the E-nullcline to obtain I_E^{ext} and count fixed points.
4. If *any* such I_E^{ext} yields a unique fixed point, the Poincaré-Bendixson theorem guarantees a limit cycle.

Stage	Count	Osc	Rate
Structural condition fails (Equation 33)	1698	—	—
I-nullcline bound fails ($\min R > w_{EI} w_{IE}$)	2223	—	—
Bound passes, but $w_{EE} F_E'$ outside window at all points	344	—	—
Unstable fixed point exists, only in multi-FP region	371	1	0.3%
Unique unstable fixed point found	364	356	97.8%

Table 8: I-nullcline sweep validation ($N = 5000$).

The results sharpen the role of uniqueness. When the sweep finds an I_E^{ext} yielding a unique unstable fixed point, oscillation is confirmed at 97.8%,which is consistent with the Poincaré-Bendixson guarantee (the 2.2% gap is attributable to oscillation detector sensitivity). In contrast, when the most favourable I_E^{ext} places an unstable fixed point in a multi-fixed-point configuration, only 0.3% oscillate (1 out of 371[10]), despite the Jacobian satisfying $\tau > 0$ and $\Delta > 0$ at the selected point. This near-zero rate persists even though the algorithm chose the I_E^{ext} maximising τ, giving the multi-fixed-point system its best chance at oscillation.

The 97.8% vs 0.3% contrast confirms that uniqueness of the unstable fixed point is not merely a technical convenience but the essential dynamical divide: the competing stable attractor in multi-fixed-point systems captures trajectories before a limit cycle can form. The 371 samples where all unstable fixed points occur in multi-fixed-point configurations represent parameter regimes where the Hopf region on the I_E^{ext} axis lies entirely within the three-fixed-point region, in which the saddle-node bifurcation occurs only after the Hopf condition has already been lost. Closing this gap analytically remains discussed in Section 5.

[9]The compact positively invariant region required by the Poincaré-Bendixson theorem is not exactly $[0, 1]^2$: the correction term $F(0) = 0$ allows $F(x) < 0$ for negative inputs, so r_E can be driven slightly below zero. A rigorous bound can be obtained by noting that $F(x) \geq -c$ where $c = \frac{1}{1+e^{a\theta}}$, giving $\dot{r}_E = 0$ when $r_E = F(u_E) \geq -c_E$. The region $[-c_E, 1 - c_E] \times [-c_I, 1 - c_I]$ is therefore positively invariant and compact, satisfying the theorem's requirements. The numerical values $c_E, c_I < 0.04$ for typical parameters, so the deviation from $[0, 1]^2$ is minor.

[10]Whether the rare multi-fixed-point oscillations (1 out of 371 in the I-nullcline sweep, 10 out of 162 in the three-level Monte Carlo) represent true supercritical limit cycles coexisting with the stable attractor or transient spirals that eventually decay would require computing the first Lyapunov coefficient at the unstable fixed point, which is a direction left for future investigation.

Comparing with the three-level Monte Carlo (Section 3.2.3), both approaches converge on the same conclusion through independent methods:

	Three-level	**I-nullcline sweep**
Method	Fixed I_E^{ext}, check levels 1–3	Sweep I_E^{ext}, find optimal
Unique unstable FP → osc	89.7%	97.8%
Multi-FP unstable → osc	6.2%	0.3%
Theoretical violations	0	0

Table 9: Comparison of validation approaches ($N = 5000$ each).

The higher rate for the I-nullcline sweep (97.8% vs 89.7%) follows from its design: instead of testing a randomly sampled I_E^{ext}, it selects the I_E^{ext} that maximises instability, producing larger limit cycles that are easier to detect. The lower multi-FP rate (0.3% vs 6.2%) reflects the same effect in reverse - giving multi-FP systems their best I_E^{ext} reveals that even under optimal conditions, the competing stable attractor almost always suppresses oscillation.

3.3 Further discussion

Our analytical conditions are consistent with the numerical bifurcation results of Li et al. (2022), who investigated the role of self-feedback loops in regulating gamma oscillations in the Wilson-Cowan model.

Their one-parameter bifurcation with respect to w_{II} (their Figure 3A) shows that increasing inhibitory self-feedback suppresses oscillation. This follows directly from the structural condition (Equation 33): the right-hand side $\frac{16\tau_E w_{II}}{\tau_I a_E}$ grows linearly with w_{II}, so for fixed cross-coupling $w_{EI}w_{IE}$, sufficiently large w_{II} will always violate the inequality. In the full master inequality (Equation 24), w_{II} enters quadratically through $(1 + w_{II}F_I')^2$, making its suppressive effect even stronger at realised fixed points.

For excitatory self-feedback, their one-parameter bifurcation with respect to w_{EE} (their Figure 6A) reveals that oscillation exists only within a limited window ($13.57 < w_{EE} < 35$). Their study concludes that "the excitatory self-feedback loop promotes the generation of gamma oscillations". However, this statement captures only the lower boundary of the window.

Since w_{EE} does not appear in the structural condition (Equation 33), its role is not to determine *whether* the network can support oscillation, which is governed by $w_{EI}w_{IE}$, w_{II}, τ_E/τ_I, and a_E. Rather, w_{EE} determines whether oscillation is *realised*, by tuning the fixed point into the feasibility window. Within this window, increasing w_{EE} does promote oscillation, it drives $w_{EE}F_E'$ above the trace threshold, consistent with Li et al.'s conclusion. But this promotion is bounded above: excessive w_{EE} either pushes u_E past θ_E into saturation (collapsing F_E', causing the trace itself to fail), or, even before saturation, drives $w_{EE}F_E'$ above the algebraic ceiling from Equation 22, producing a Saddle. The complete picture is therefore that w_{EE} acts as a *tuning parameter* with a finite effective range, not a monotonic promoter of oscillation.

We refer to the following papers:

Inhibitory self-feedback. Li et al. (2022):

> The present results reveal that, on one hand, the inhibitory self-feedback loop is not conducive to the generation of gamma oscillations, and increased inhibitory self-feedback strength facilitates the enhancement of the oscillation frequency. On the other hand, the excitatory self-feedback loop promotes the generation of gamma oscillations [...] Inhibitory and excitatory self-feedback loops play a complementary role in generating and regulating the gamma oscillation in Wilson-Cowan model.

Time constants. Keeley et al. (2019):

> The models' flexibility to capture the broad range of gamma behavior depends directly on the timescales that represent recruitment of the excitatory and inhibitory firing rates. In particular, we find that weakly modulated gamma oscillations occur robustly when the recruitment timescale of inhibition is faster than that of excitation.

4 Limitations

Several limitations of the present analysis should be noted.

Decoupled optimisation. The structural condition is obtained by independently maximising F_E' and minimising the cost over F_I'. In the actual system, both derivatives are jointly determined by a single fixed point on the nullcline intersection, and may not simultaneously attain their optimal values. This makes Equation 33 a potentially loose bound. The I-nullcline analysis (Section 3.1.2) partially closes this gap by coupling the derivatives along the I-nullcline, showing that the decoupled optimum is an exact stationary point of the coupled problem. However, the remaining distance to a closed-form sufficient condition persists (see Section 5). Additionally, the optimal $F_I' = 1/w_{II}$ may exceed the sigmoid maximum $a_I/4$. When $w_{II} < 4/a_I$, the true minimum of the right-hand side is attained at the boundary $F_I' = a_I/4$, making Equation 33 looser than necessary. The condition remains valid but could be tightened to $w_{EI}w_{IE} > \frac{16\tau_E(1+w_{II}a_I/4)^2}{\tau_I a_E a_I}$ in this regime.

Finite numerical scope. The Monte Carlo study ($N = 5000$) found no three-fixed-point system with two unstable equilibria (U2/S0/Sad1), but this is an empirical observation, not a topological proof. The Poincaré index theorem permits a limit cycle enclosing all three such fixed points (index sum $= +1$), so this configuration cannot be ruled out in other parameter regimes. Similarly for the outstanding data in Section 3.2.3, we did not mathematically confirm it is impossible.

Sigmoid form. All results assume the specific sigmoid with the correction term $F(0) = 0$. The original Wilson-Cowan formulation includes a refractory term $(1 - r_E)$ which modifies the effective gain; our conditions do not directly apply to that variant.

Single-population time constants. The model assumes each population is characterised by a single time constant. Distributed delays or synaptic filtering (as in Keeley et al. (2019)) may shift the oscillatory boundaries quantitatively, though the qualitative structure of the trace-determinant framework is preserved.

5 Future work

The I-nullcline sweep (Section 3.2.4) provides a numerical algorithm for determining whether oscillation-producing I_E^{ext} exists, but the analytical question remains open: can the overlap between the Hopf region and the one-fixed-point region on the I_E^{ext} axis be characterised in closed form?

The key geometric ingredient is the E-nullcline fold. The trace condition requires $w_{EE}F_E' > 1$, which at the sigmoid midpoint demands $w_{EE} > 4/a_E$. But this is precisely the condition for the E-nullcline to be non-monotone (S-shaped), guaranteeing multiple fixed points at some input levels. As I_E^{ext} increases, the E-nullcline shifts upward until a saddle-node bifurcation annihilates two fixed points, leaving a unique equilibrium. If this surviving fixed point satisfies the Hopf condition, a limit cycle is guaranteed by the Poincaré-Bendixson theorem.

The question therefore reduces to whether the Hopf bifurcation occurs in the one-fixed-point region of the $(I_E^{\text{ext}}, w_{EE})$ parameter plane. Completing this analysis would close the gap between structural necessity (Equation 33) and a fully parametric sufficient condition, eliminating the need for fixed-point enumeration.

6 Key References

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